Combinational Circuits

1. In sequential circuits, the outputs depend on……
   A) The inputs only
   B) The inputs and the previous states
   C) The previous states only
   D) None of the above

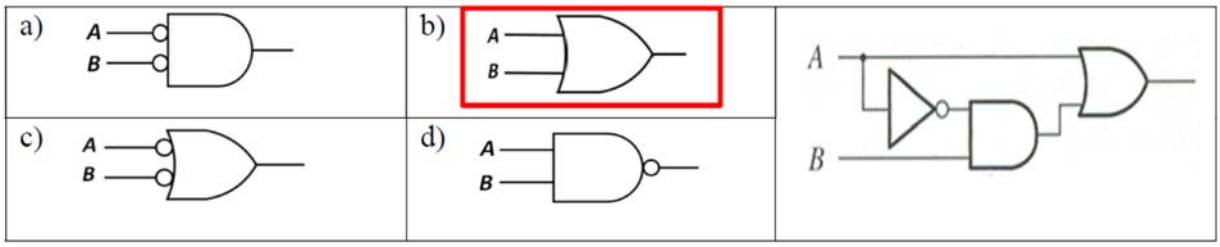
2. Which of the following do you need to construct an 8-bit binary adder?
   A) Eight full adders
   B) Eight half adders
   C) one full adders and three half adders
   D) two full adders and two half adders

3. ……..selects binary information from many input lines and directs it into a single output line.
   A. Multiplexer B. Encoders C. Decoders D. Demultiplexer

4. When two values of the same sign are added and the result have opposite sign, this is called?
   a) underflow b) overflow c) signflow d) None of the above

5. When the control input disables a tri-state gate, the output will be ------.
   a) 0 b) 1 c) Hi-Z d) None of the above

6. The circuit shown in the following figure is equivalent to.



7. The circuit in figure 4 is equivalent to the circuit in ----------.

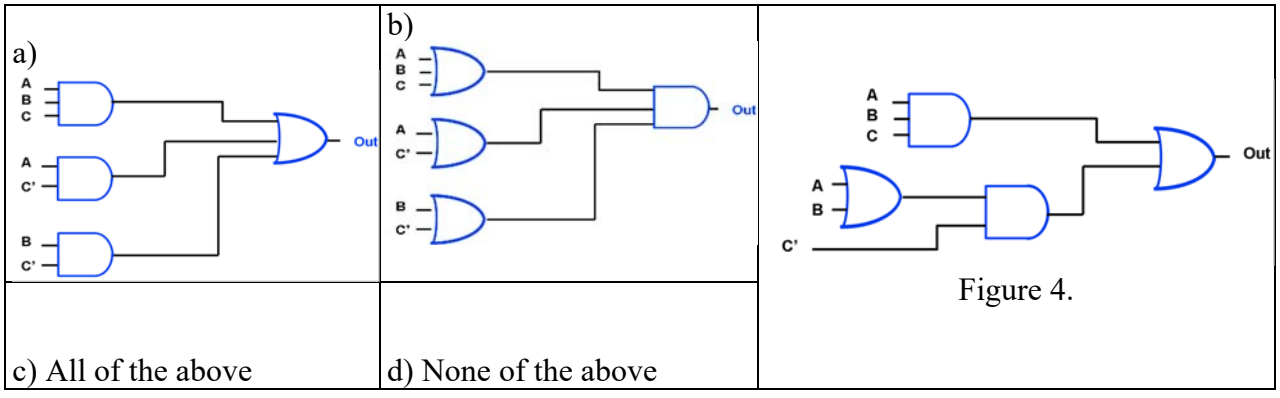


Figure 4.

c) All of the above

d) None of the above

8. Use 4x1 Multiplexer to implement the Boolean function $f(x, y, z) = \sum(0, 2, 3, 5)$.

9. Using a decoder and external gates, design the combinational circuit defined by the following three Boolean functions:

   $F1(A, B, C) = \Sigma(2, 4, 7)$
   $F2(A, B, C) = \Sigma(0, 3)$